

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 03)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1 . 1**

Which of the following is a feature of natural systems? Tick **one** box.

**[1 mark]**

- A Reactions that require high temperatures ☐
- B Reliance on non-renewable energy ☐
- C Waste products that are the raw materials for other processes ☐
- D Dispersal of elements so they cannot be re-used ☐

**0 1 . 2**

Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

**[2 marks]**

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**0 1 . 3**

Explain why incineration of wastes is less sustainable than natural decomposition.

**[2 marks]**

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**5**

**0 2 . 1**

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

**[4 marks]**

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02 . 2

Explain why growing coral reefs help to resist climate change.

[2 marks]

02 . 3

Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

Do not write outside the box

Extra space

15

03 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1 .....

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2 .....

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03 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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03 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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5

04

Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

| Component of catch                        | Mass / tonnes |
|-------------------------------------------|---------------|
| Shrimps (target species)                  | 0.6           |
| Crabs, sea urchins, molluscs and starfish | 5.9           |
| Immature fish of commercial species       | 1.9           |

04 . 1

Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer

%

0 4 . 2

Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

0 4 . 3

Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

0 4 . 4

Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

[illegible]

15

**0 5** Table 1 shows some designated protected areas in the UK.

**0 5** . **1** Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

| Designation                                | Designated under | Key feature |  |
|--------------------------------------------|------------------|-------------|--|
| Site of Special Scientific Interest (SSSI) |                  |             |  |
| Special Protection Area (SPA)              |                  |             |  |
| Ramsar site                                |                  |             |  |
| Marine Conservation Zone (MCZ)             |                  |             |  |

**0 5** . **2** Explain how designating a protected area protects habitats and species.

[3 marks]

05 . 3

Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

10

06 . 1

Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

06 . 2

Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

06 . 3

Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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5

07 . 1

Explain the effects of deforestation on soil.

[4 marks]

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07 . 2

Explain how deforestation affects hydrology and climate.

[4 marks]

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07 . 3

Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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1 .....

2 .....

10

0 8 . 1 Define each of the following ecological terms and give an example.

[8 marks]

**Species** .....

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**Ecological niche** .....

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**Community** .....

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**Biome** .....

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0 8 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

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10

09 . 1

Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

09 . 2

For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

**Introduced competitor** .....

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**Introduced predator** .....

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**Introduced pathogen** .....

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09 . 3

Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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10

10 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

Extra space

10 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

10

Do not write outside the box

**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

**[25 marks]**

**OR**

**1 1 . 2**

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

[illegible]

END OF QUESTIONS